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Summary

Edge-AI inference engineer who owns **both halves of the stack** — the **DeepStream / TensorRT / CUDA** computer-vision pipeline and the **vLLM**-served VLM/LLM layer that reasons over it. Core engineer on a production video-analytics platform running across **10+ deployment sites**; I turn detection accuracy, GPU throughput, and reliability into **measured** wins. IIT Bombay, CSE.

Experience

Software Development Engineer <i>Avathon (formerly SparkCognition)</i>	Jul 2025 – Present Bengaluru, India
• Core engineer on the ground-up rewrite of IRIS, Avathon’s production video-analytics platform (10+ sites), from DeepStream 6.2 → 9.0; designed the SDK/bundler split with a stable plugin ABI — self-registering verticals, 3 shared libraries loading 11 use-cases — so the C++/GStreamer core, MQTT/cloud integration, and use-case modules ship independently. • Root-caused a total detection failure in the DEIMv2 (DETR) detector to two independent bugs: a TensorRT 10.14 miscompile of GridSample-based deformable attention (isolated via polygraphy layer bisection, fixed with ONNX graph surgery) and a silent PtxVersion(222) JIT failure feeding the engine zeros — restoring 6,000+ detections. • Cut detector latency 3.3× (15.5 → 4.7 ms/frame) with TensorRT FP16, and migrated the tracker NvDCF → NvSORT for a 7× faster tracking stage (2.6 → 0.37 ms p50) rolled out across 32 production configs; authored custom C++ GStreamer plugins with zero-copy NvBufSurface buffers and NV12→BGR conversion. • Nearly tripled per-GPU camera capacity (~2.7×) via a per-camera activity gate (+40–100% fps on busy cameras, –35% end-to-end p50 latency); benchmarked and rejected Triton with data (nvinfer 178 vs nvinferserver 66 fps) and proved MIG/MPS/INT8 dead levers on a compute-saturated GPU. • Architected VSS, a vLLM-served VLM+LLM layer (Cosmos-Reason2-2B) over Kafka/Redis for real-time scene understanding and alert verification; fed GPU-resident image embeddings straight into vLLM to skip the CPU pre-processing round-trip — 4.3× throughput (42 → 180 tok/s) for +40 MiB, with byte-identical output parity. • Debugged fleet-wide systems failures: cross-version memory leaks (DS6.2/DS8 GLib-managed vs. raw malloc), ABI / ODR symbol-interposition crashes, and RTSP reconnect wedges — decoupling TensorRT engine batch-size from camera count and adding watchdogs so the fleet is stable regardless of size.	
Software Developer Intern <i>Hilti Technology Solutions India</i>	May 2024 – Jul 2024 Pune, India
• Automated MSI installer creation via a GitLab CI/CD pipeline (Docker + WiX Toolset CLI), cutting deployment from 2 hours to 10 minutes; built Selenium (C#) web UI test automation with the Page Object Model.	

Projects

Kirn — AI Terminal Python, Ollama, Local LLMs	Open Source
• A real bash shell with an inline local-LLM assistant — tool-calling, autonomous command execution, PTY support, and automatic error explanation; runs on Linux and Android/Termux (~1,300 LOC).	
Church Live Translation Python, faster-whisper, IndicTrans2, WebSocket	
• Offline real-time speech-to-caption translation: streaming Whisper ASR (custom LocalAgreement) → batched EN→ML/TE/HI neural translation → WebSocket broadcast, all within ~3–4 GB VRAM on a single RTX 3060.	
Semi-Supervised Skin-Lesion Classification PyTorch, ResNet-50	Aug 2023 – Nov 2023
• MixMatch pipeline for 7-class classification over 10k dermatoscopic images using pseudo-labelling, consistency regularisation, and mixup. <i>Guide: Prof. Preethi Jyothi, IIT Bombay.</i>	

Technical Skills

Languages	Python, C++, C, Bash, SQL
Inference & Edge	NVIDIA DeepStream, TensorRT, CUDA, GStreamer, Triton, ONNX Runtime, FP16/INT8 Quantization
LLM & ML	vLLM, PyTorch, VLMs/LLMs, Hugging Face, OpenCV, NumPy, Pandas, Scikit-learn
Systems & Infra	Docker, Kubernetes, Kafka, Redis, MQTT, Linux, Git, GitLab CI/CD, meson/ninja

Education & Achievements

Indian Institute of Technology Bombay <i>Bachelor of Technology in Computer Science and Engineering</i>	2021 – 2025
JEE Advanced AIR 709 · JEE Main AIR 157 (top 0.02% of 1.2M) · NTSE Scholar (NCERT, Govt. of India)	